Dynamic Treatment Regimes

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Dynamic Treatment Regime

- Goal here is to extend what we have learned about (optimal) treatment regimens for single stage decisions to multiple decisions
- Use similar notation as we have introduced previously for multiple decision points
- First discuss why they are useful, formally define a DTR, describe how we can estimate and test different components of a DTR

Motivation for Dynamic Treatment Regime

- Not everyone responds to the same intervention in the same way (between person heterogeneity)
- Not everyone response to an intervention is consistent over time (within person heterogeneity) due to changes in adherence, engagement, burden, or accrual of toxicity or delayed side effects
- Many chronic or long-term conditions or diseases -cancer, smoking, substance use disorder, depression, obesity, ADHD, autism, or schizophrenia - cannot be cured with a single treatment
- BUT we may have multiple different interventions (or dosages/intensities to try)

How do we treat chronic disease?

- Make an initial treatment decision, monitor for response/adherence, and adjusting treatment based on the individual's changing course
- Can we formalize this process to test different strategies improve outcomes?

Dynamic Treatment Regime

- Pre-specified sequence of decision rules
- One rule for each possible decision point
- Dictate what type of treatment/intervention, at what dosage/intensity, with what type of delivery (e.g., switch, augment, intensify, stay the same)
- Based on measured patient characteristics (including prior response to treatment)

Dynamic Treatment Regime Formally

- Sequence of functions $g_0, \ldots, g_K$, one for each possible decision point
- Which map from the sample space of treatment and covariate history to the set of possible treatments at that particular decision point
- Note that the set of possible treatment at any particular may depend on past treatment and covariate history

Dynamic Treatment Regimes

- The intervention/treatment strategy/regime/policy is said to be "adaptive" or "dynamic" or "tailored" because it depends on a person's evolving characteristic
- Note: Stepped care intervention models are a special case of adaptive interventions

A Rose By Any Other Name . . .

- Adaptive Intervention Strategy (AI or AIS)
- Adaptive Treatment Strategy (ATS)
- Dynamic Treatment Regime (DTR)
- Multistage Treatment Strategy (MTS)
- Treatment Policy (TP)

Example DTR in Smoking Cessation

- Stage 1: Initial Phase TLC (phone counseling + NRT) for 8 weeks
- Stage 2: At end of 8 weeks assess 7 day abstinence IF abstinent (complete responder) continue with TLC (at least monthly phone counseling + NRT) for 48 weeks ELSE combine TLC with pharmacotherapy (MTM)

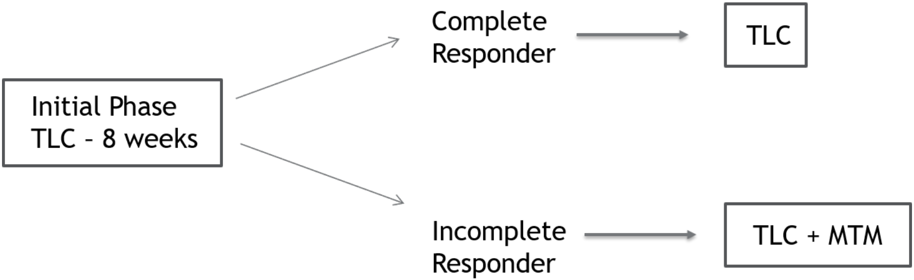

Figure 1: Smoking Cessation DTR

Tailoring Variable

- The variable and cutoff which determine which intervention a participant should receive are known as tailoring variables and cutpoint
- Smoking cessation DTR: 7 day abstinence
- What can be a tailoring variable?
- Any variable which can be measured can be a tailoring variable
- This does not mean all are equally good ideas

Many possible DTRs

- Typically used the following to develop DTR - (a) Expert opinion, (b) Clinical judgement, (c) Piecing together an adaptive intervention using results from separate RCTs
- Are these "good" adaptive interventions?
- How could these adaptive interventions be improved?

Unanswered Questions to Optimize a DTR

- Which treatment or intervention should be offered to individuals initially?
- How frequently should individuals be monitored for response/nonresponse to initial treatment?
- Among individuals who are not responding well to initial treatment, should their initial treatment be intensified, augmented, or switched?
- Among individuals responding to treatment, what maintenance/relapse prevention treatment should be offered?

Expected Response Under a Dynamic Treatment Regime

- Suppose I want to estimate the anticipated response if all subjects were to follow the dynamic treatment regime $g_0(L_0), g_1(\overline{L}_1, A_0), \ldots, g_K(\overline{L}_K, \overline{A}_{K-1})$

1. Regression-based methods (extension of G-computation algorithm)
2. Inverse probability weighted methods (extension of IPW estimators)

(Parametric) G-computation Algorithm

- To do the (Parametric) G-computation algorithm in practice we could proceed as follows.

1. Posit models for the conditional densities (this is exactly the same as before when we estimated the effect of static regimes)

$P_{L_j|\overline{L}_{j-1},\overline{A}_{j-1}}(l_j|\bar{l}_{j-1},\overline{a}_{j-1};\psi_j)$ for $j = 1,\ldots,M$

$P_{L_0}(l_0;\psi_0)$

$P_{Y|\overline{L}_M,\overline{A}_M}(y|\bar{l}_M,\overline{a}_M;\psi_{M+1})$

in terms of the parameters $(\psi_0,\ldots,\psi_{M+1})$ where ψ_j may be a vector of parameters. Note that some of these parameters could be shared across different time points

(Parametric) G-computation Algorithm

2) Obtain parameter estimates using least squares/maximum likelihood using standard software

3) Integrating out the $L's$ analytically would likely be prohibitive. Therefore, we could approximate the distribution of $Y^{\overline{a}}$ for any $\overline{a}$ of interest using Monte Carlo integration. Specifically, for $r = 1, \ldots, K$ (number of simulations), fix $\overline{a}$ and then

i) generate a random l_{0r} from $P_{L_0}(l_0; \hat{\psi}_0)$

ii) generate a random l_{1r} from $P_{L_1|L_0,A_0}(l_1|l_{0r}, g_0(l_{0r}); \hat{\psi}_1)$

iii) generate a random l_{jr} from $P_{L_j|\overline{L}_{j-1},\overline{A}_{j-1}}(l_j|\bar{l}_{j-1,r}, \overline{g}_j(\bar{l}_{jr}, \overline{g}_{j-1}); \hat{\psi}_j)$ for $j = 2, \ldots, M$

iv) generate a random y_r from $P_{Y|\overline{L}_M,\overline{A}_M}(y|\bar{l}_{M,r}, \overline{g}_M; \hat{\psi}_{M+1})$

(Parametric) G-computation Algorithm

- Note that as I am simulating the covariate and outcome history on each subject, I am assigning treatment based on the treatment rule that I am interested in studying (e.g., $g_j()$) and not the same treatment for each subject (as we had done in the past)

(Parametric) G-computation Algorithm

- The y_r for $r = 1, \ldots, K$ derived this way represent random draws from the marginal distribution of $Y^{\overline{g}}$.
- For example, we could estimate $E\{Y^{\overline{g}}\}$ by $\frac{1}{K}\sum_{r=1}^{K} y_r$
- We are then able to compare different treatment combinations in this fashion
- Standard errors could be estimated using the bootstrap. Note that there are two "Monte Carlo" steps here - for the bootstrap and the integration

HIV Example

- Consider the following dynamic treatment regime: "Give treatment if the current CD4 count falls below x as long as the patient has not become resistant; otherwise do not treat."
- Use the g-computation algorithm estimate the mean response for such a dynamic treatment regime for x=200,300,400
- Code is very similar to the one of static regimes except for the lines for treatment allocation

G-computation Implementation

```
mean_response <- function(threshold, coef_estimates, MC) {
    mu0 <- coef_estimates[1]; sigma0 <- coef_estimates[2]
    beta0 <- coef_estimates[3]; beta1 <- coef_estimates[4]; be
    beta3 <- coef_estimates[6]; beta4 <- coef_estimates[7];
    beta5 <- coef_estimates[8]; beta6 <- coef_estimates[9];
    beta7 <- coef_estimates[10];

    sigma <- coef_estimates[11];gamma0 <- coef_estimates[12];g

        R0 <- rep(0, MC)
        logCD40 <- rnorm(MC, mu0, sigma0)
        A0 <- ifelse(exp(logCD40) < threshold & R0 == 0, 1, 0)
        cumA0 <- A0
        R1 <- rbinom(MC, 1, expit(gamma0 + gamma1*cumA0))*(cu
            0*(cumA0 == 0) + 1*(R0 == 1)
        logCD41 <- beta0 + beta1*logCD40 + beta2*A0 + beta3*R1
            beta4*logCD40*A0 + beta5*logCD40*R1 + beta6*A0*R1
```

Estimated Average CD4 Under Different Dynamic Treatment Regimes

```
##                    mean_logCD4 mean_CD4
## Threshold = 200           5.69      364
## Threshold = 300           5.80      408
## Threshold = 400           5.86      448
```

Standard Errors of Estimated Average CD4 Under Different Dynamic Treatment Regimes

```
##                   SE_logCD4 SE_CD4
## Threshold = 200       0.030   11.7
## Threshold = 300       0.022   10.4
## Threshold = 400       0.020    9.8
```

Inverse Probability Weighting

- Instead of simulating the entire covariate trajectory, could take a weighted average of those subjects whose covariate history is consistent with the regime of interest
- We can derive a consistent estimator for $E(Y^{\overline{g}})$ by using

$$n^{-1} \sum_{i=1}^{n} \frac{\prod_{j=0}^{M} I(A_{ij} = g_j(\overline{L}_{ij}, \overline{A}_{i,j-1}))}{\prod_{j=0}^{M} P_{A_j|\overline{A}_{j-1},\overline{L}_j}(A_{ij}|\overline{A}_{i,j-1}, \overline{L}_{ij})} Y_i \tag{1}$$

HIV Example

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Denominator in IPW

```r
# calculate the probability of receiving treatment (i.e. A = 1) and
# probability the participant received treatment they actually did
dataHW3_long <- mutate(dataHW3_long,
    pred_trt = predict(m3, newdata = dataHW3_long, type = "response"
    pred_trt_received = (pred_trt^A)*(1-pred_trt)^(1-A))

# calculate denominator weights
dataHW3$den_wt <- tapply(dataHW3_long$pred_trt_received,
    dataHW3_long$ID, prod, na.rm = TRUE)
```

Numerator in IPW

```
mean_response <- function(threshold) {
    dataHW3 <- mutate(dataHW3,
        # find which treatment a subject was supposed to receive
        # based on the regime
        A0_regime = ifelse(CD40 < threshold & R0 == 0, 1, 0),
        A1_regime = ifelse(CD41 < threshold & R1 == 0, 1, 0),
        A2_regime = ifelse(CD42 < threshold & R2 == 0, 1, 0),
        A3_regime = ifelse(CD43 < threshold & R3 == 0, 1, 0),
        # find out whether or not they were compliant
        compliant0 = ifelse(A0 == A0_regime, 1, 0),
        compliant1 = ifelse(A1 == A1_regime, 1, 0),
        compliant2 = ifelse(A2 == A2_regime, 1, 0),
        compliant3 = ifelse(A3 == A3_regime, 1, 0),
        # get numerator weight
        num_wt = compliant0*compliant1*compliant2*compliant3,
        # overall weight
        wt = num_wt/den_wt
    )
    mean_response <- mean_response <- c(
        weighted.mean(dataHW3$logCD44, dataHW3$wt)
```

Estimated Average CD4 Under Different Dynamic Treatment Regimes

```
##                    mean_logCD4 mean_CD4
## Threshold = 200           5.68      349
## Threshold = 300           5.80      393
## Threshold = 400           5.77      404
```

Standard Error of Estimated Average CD4 Under Different Dynamic Treatment Regimes

```
##                    SE_logCD4 SE_CD4
## Threshold = 200       0.060     23
## Threshold = 300       0.067     24
## Threshold = 400       0.077     29
```